

Working Title: A Study of Low-Light Enhancement in the HVI Color Space

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Abstract

HVI-CIDNet (Yan et al., 2025a) enhances low-light images by mapping them into the HVI (Horizontal/Vertical/Intensity) color space, processing intensity and chroma along separate branches, and supervising the output in both RGB and HVI. This thesis studies a question about that design; the question, the claims, and the experiments are fixed in the project plan before any run starts, and every reported number is read from a result file by a verification script rather than transcribed.

1. Introduction

Low-light image enhancement recovers a well-exposed image from an underexposed capture. Learned enhancers now dominate the paired benchmarks (Wei et al., 2018; Cai et al., 2023; Guo et al., 2020), and a recent line of work argues that the color space in which the network operates and is supervised matters as much as the architecture. HVI-CIDNet (Yan et al., 2025a) maps images into a polar Horizontal/Vertical/Intensity (HVI) space, enhances intensity and chroma along separate interacting branches, and is trained with a loss summed over both the RGB output and its HVI transform.

This paper asks one question about that design. The question, the claims it supports, and the experiments that decide it are stated in Section 2; the results follow in Section 3.

2. Method and Protocol

Model. We use the public HVI-CIDNet implementation (Yan et al., 2025a). The network takes an sRGB image, converts it to HVI, processes the intensity map and the chroma map along two interacting branches, and converts the result back to sRGB.

Protocol. All runs share one frozen protocol: the same dataset split, seeds, optimizer, schedule, and hardware (a

single NVIDIA GeForce RTX 2080 Ti (11 GB) in FP32). Checkpoints are selected on the validation split only; the test split is read once, by a separate final-evaluation script, per reported configuration. Each run records the SHA-256 of its training script, the upstream commit, and the split file.

3. Results

Results are reported here once the studies defined in the project plan have run. Every table in this section is generated from a result file, and every number is bound to the key it was read from.

4. Conclusion

We study one question about the HVI-CIDNet design under a single frozen protocol, and report every finding against its seed spread. The protocol and scripts are released with the thesis.

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A. Related Work

Retinex-based enhancement. Retinex theory (Land, 1977) models an image as the product of illumination and reflectance, and most learned low-light enhancers descend from it. RetinexNet (Wei et al., 2018) learns the decomposition end to end and introduced the LOL-v1 benchmark; KinD (Zhang et al., 2019) enhances illumination and reflectance along separate branches; URetinex-Net (Wu et al., 2022) and RUAS (Liu et al., 2021) unroll the decomposition into a network, the latter with architecture search. Retinexformer (Cai et al., 2023) folds the illumination estimate into a one-stage transformer and is the published model closest in size to HVI-CIDNet.

Supervised enhancers in other formulations. LLFlow (Wang et al., 2022) models the conditional distribution of normal-light images with a normalizing flow; SNR-Aware (Xu et al., 2022) routes between local and global processing by a spatial signal-to-noise estimate; LLFormer (Wang et al., 2023) targets ultra-high-definition inputs; GSAD (Hou et al., 2023) and Diff-Retinex (Yi et al., 2023) use diffusion models. The channel-attention block inside HVI-CIDNet follows the design of Restormer (Zamir et al., 2022). LEDNet (Zhou et al., 2022) treats enhancement and deblurring jointly and introduced the LOL-Blur dataset.

Unsupervised and zero-reference methods. Zero-DCE (Guo et al., 2020) estimates per-pixel enhancement curves without paired data; EnlightenGAN (Jiang et al., 2021) trains adversarially on unpaired images; SCI (Ma et al., 2022) learns a self-calibrated illumination map with a very small network. HVI-CIDNet is trained with pairs only and its authors list unsupervised training as open.

Color spaces for enhancement. Most of the methods above operate on sRGB. LYT-Net (Brateanu et al., 2025) processes luminance and chrominance separately in YUV. HVI-CIDNet (Yan et al., 2025a) instead maps sRGB into a polar Horizontal/Vertical/Intensity (HVI) space in which hue is an angle, saturation a radius, and the radius is shrunk toward the origin as intensity goes to zero by a learned collapse function; the network enhances intensity and chroma along two interacting branches and is supervised in both sRGB and HVI. An earlier version of the method (Yan et al., 2024) and a successor for extreme darkness (Yan et al., 2025b) exist; we study the CVPR 2025 version whose code is public.

Datasets and metrics. We use the paired LOL-v1 benchmark of Wei et al. (2018); its successor LOL-v2 (Yang et al., 2021), SID (Chen et al., 2018), SICE (Cai et al., 2018), and LOL-Blur (Zhou et al., 2022) are the other paired sets on which HVI-CIDNet was reported. Full-reference quality is measured by PSNR, SSIM (Wang et al., 2004), and LPIPS (Zhang et al., 2018); no-reference quality by NIQE (Mittal et al., 2013). The perceptual loss term of HVI-CIDNet is the VGG feature loss of Johnson et al. (2016). Color fidelity, when we report it, uses the CIEDE2000 difference (Sharma et al., 2005).

B. Additional Experiments

Full tables, per-image analysis, and protocol details are reported here once the studies have run.

B.1. Protocol Details

Each run directory records the training-script SHA-256, the upstream HVI-CIDNet commit, the split-file SHA-256, the torch version, and the GPU name. The validation summary written at the end of every run is the only source of the numbers in this paper.